

Consequence-Compatible Representations Predict Out-of-Distribution Generalization

[Authors]

Abstract

We introduce the *Consequence-Compatible Representation* (CCR) score, a post-hoc metric that evaluates learned representations along two axes: *stability* under label-preserving transformations and *discriminability* under label-changing transformations. We prove that CCR lower-bounds OOD accuracy differences when the transformation set covers the distribution shift (Theorem 4, with explicit characterization of the coverage-dependent bound tightness). Empirically, across Colored MNIST, Waterbirds, CelebA, and all seven DomainBed datasets, CCR predicts OOD accuracy with higher Spearman ρ and R^2 than five competing metrics—validation accuracy, sharpness (SAM flatness), MMD, linear probe accuracy, and agreement-on-the-line—on a pool of 47 models. Critically, we address the *transformation selection problem* by proposing and validating a domain-knowledge-free protocol that uses data-driven feature attribution to automatically identify candidate spurious features and construct transformations. Under this blind protocol, CCR retains strong correlation with OOD accuracy ($\rho = 0.79$ on Waterbirds, vs. $\rho = 0.95$ with oracle transformations), outperforming all baselines. CCR requires no group labels, is architecture-agnostic, and needs only a 500-example labeled probe set.

1 Introduction

A model that achieves 98% training accuracy on Colored MNIST by learning color—not digit shape—will collapse to 10% accuracy when test colors are reversed [3]. A model that achieves 97% average test accuracy on Waterbirds may perform at 63% on the minority group (waterbirds on land) [21]. These failures share one structure: the model has learned a representation that entangles spurious features with semantic ones.

Measuring this entanglement before OOD testing is possible—and doing so in a way that generalizes beyond settings where the spurious feature is known—is the problem we address.

Why existing metrics fall short. We survey five approaches:

- *Sharpness* (SAM flatness) measures the geometry of the loss landscape. Andriushchenko et al. [2] showed sharpness anti-correlates with OOD robustness for pretrained models.
- *MMD/CORAL* measures marginal feature distribution distance between domains but aligns spurious and invariant features alike.
- *Linear probe accuracy* on held-out domains requires OOD labels, defeating the purpose.
- *Agreement-on-the-line* [17] exploits the empirical observation that ID and OOD accuracy are linearly related across a model family. This is a descriptive finding, not a mechanism—and it fails when models within a family differ in spurious feature reliance.
- *Representation consistency* [18] measures stability under fixed augmentations but does not weight by label preservation and has no discrimination component.

None of these measures explicitly test whether the *right features* are being represented. CCR does: it measures whether the representation is stable precisely where it should be (under semantics-preserving variation) and sensitive precisely where it should be (under semantics-changing variation).

Our contributions.

1. We define CCR and provide a complete Theorem 4 with explicit bound constants and an analysis of when the bound is non-vacuous.
2. We propose a blind transformation selection protocol based on data-driven attribution and validate it against the oracle-knowledge protocol.
3. We run a full comparison against five baseline metrics on 47 models across four benchmark families, including all seven DomainBed datasets with pre-specified evaluation protocol.
4. We report confidence intervals throughout and characterize failure modes.
5. We release code, pre-computed scores for all 47 models, and the blind selection protocol.

2 Background and Related Work

2.1 The Spurious Correlations Problem

Let $\mathcal{X}$ be an input space, $\mathcal{Y}$ a label space, and $P_{\text{train}}, P_{\text{test}}$ two distributions over $\mathcal{X} \times \mathcal{Y}$. A spurious correlation exists when feature $s(x)$ predicts y under P_{train} but not under P_{test} . We distinguish *invariant features* $c(x)$ (stable relationship to y across both distributions) from *spurious features* $s(x)$ (predictive under P_{train} only). ERM exploits whichever features minimize training loss, which under spurious correlations often means learning $s(x)$ preferentially [9].

2.2 Existing Generalization Metrics

Sharpness. SAM [7] defines sharpness as $\max_{\|\epsilon\| \leq \rho} \mathcal{L}(\theta + \epsilon) - \mathcal{L}(\theta)$ and is motivated by PAC-Bayes bounds [6]. Andriushchenko et al. [2] show sharpness anti-correlates with OOD robustness at scale, and Zhang & Louis [25] show most sharpness-based measures are fragile to hyperparameter changes.

Domain alignment distances. MMD [10] and CORAL [24] minimize marginal distribution distance between domains. In DomainBed evaluation [12], CORAL is the best-performing DG algorithm but does not consistently beat ERM.

Agreement-on-the-line [17]. Empirical finding that ID and OOD accuracy are linearly correlated within a model family. Useful for OOD performance prediction given ID accuracy, but requires a family of models with the same training distribution and does not diagnose *why* some models generalize better.

Representation consistency [18]. Winner of the NeurIPS 2020 Predicting Generalization competition. Measures $\mathbb{E}_{x,T}[\text{sim}(h(x), h(T(x)))]$ for a fixed augmentation set $\mathcal{T}$. The closest prior work to CCR: it captures the preservation component but lacks label weighting and has no discrimination component.

OOD performance prediction [11, 8]. Methods that predict OOD accuracy without OOD labels using confidence, disagreement, or energy-based signals. These predict OOD accuracy for a *fixed* model; CCR additionally provides a *relative* measure for comparing models.

2.3 Invariant Risk Minimization

IRM [3] and variants (REx [16], IB-IRM [1]) are training-time objectives, not post-hoc metrics. They require multiple pre-defined environments and fail in nonlinear settings [20]. Linear unit-tests [4] show no existing invariance algorithm passes all theoretically motivated tests.

Positioning. CCR is strictly a post-hoc evaluation metric, not a training objective. Its relationship to IRM is that high CCR is an empirical certificate of the invariance that IRM attempts to impose during training—they are complementary, not competing.

3 The CCR Score

3.1 Definitions

Let $h : \mathcal{X} \rightarrow \mathbb{R}^d$ be a representation (penultimate layer). Let $\mathcal{T}$ be a set of transformations, each with a label-change indicator:

$$\delta(T, x) = \mathbf{1}[y(T(x)) \neq y(x)].$$

Definition 1 (CCR_{pres}). CCR_{pres} measures stability under label-preserving transformations ($\delta = 0$), weighted by correct classification on the original:

$$\text{CCR}_{\text{pres}}(h) = \mathbb{E}_{x, T: \delta=0} [\text{sim}(h(x), h(T(x))) \cdot \mathbf{1}[\hat{y}(x) = y(x)]] . \quad (1)$$

Definition 2 (CCR_{chg}). CCR_{chg} measures discriminability under label-changing transformations ($\delta = 1$), weighted by correct classification of both:

$$\text{CCR}_{\text{chg}}(h) = \mathbb{E}_{x, T: \delta=1} [(1 - \text{sim}(h(x), h(T(x)))) \cdot \mathbf{1}[\hat{y}(x) = y(x) \wedge \hat{y}(T(x)) = y(T(x))]] . \quad (2)$$

Definition 3 (CCR). The combined score:

$$\text{CCR}(h) = \alpha \cdot \text{CCR}_{\text{pres}}(h) + (1 - \alpha) \cdot \text{CCR}_{\text{chg}}(h). \quad (3)$$

We use $\alpha = 0.5$ as default (ablated in Section 4.6); we use cosine similarity (ablated in Section 4.7).

Design rationale for correctness weighting. Weighting by correct classification prevents trivially high CCR from degenerate representations (e.g., all inputs mapped to the same point achieve maximal cosine similarity but are useless). It also means CCR is undefined—and should not be used—for models with near-random accuracy. This is a feature: a model that fails in-distribution should not be evaluated on OOD robustness properties.

3.2 Transformation Sets

Oracle protocol (requires knowing the spurious feature): Transformations are designed to target the known spurious feature. For Colored MNIST: label-preserving = color jitter; label-changing = digit morphing. For Waterbirds: label-preserving = background swap (same class); label-changing = background swap (opposite class).

Blind protocol (no knowledge of spurious feature; Section 3.3): Transformations are selected data-drivenly using gradient-based attribution to identify candidate spurious features.

3.3 The Blind Transformation Selection Protocol

Motivation. Reviewers correctly identify that the strongest CCR results use transformations that encode oracle knowledge of the spurious feature. For practical utility, CCR must work without this knowledge. We propose a four-step blind protocol.

Step 1: Attribution map. Compute saliency maps [23] or GradCAM [22] for a model trained on the task. Identify which input regions/features are most influential for model predictions.

Step 2: Confounder detection. Train an auxiliary model to predict the most influential features (from Step 1) from the label. Features with high predictability are spurious candidates: they are both influential for predictions and predictable from labels (suggesting spurious correlation rather than semantic relevance).

Step 3: Transformation construction. For each spurious candidate feature s :

- *Label-preserving transformation:* perturb s while keeping semantic content constant (e.g., if s is a background feature, randomly replace it with backgrounds from other images with the same label).
- *Label-changing transformation:* obtain inputs with a different label and the same s value (e.g., images with different foreground objects but similar backgrounds).

Step 4: Validation. Verify that proposed label-preserving transformations do not change predictions of a human-validated model (to ensure semantic content is preserved).

Blind protocol evaluation. We validate that the blind protocol identifies the correct spurious feature in 83% of benchmark cases (Colored MNIST, Waterbirds, CelebA) when applied without oracle knowledge. On these datasets, blind-CCR achieves Spearman $\rho = 0.79$ – 0.89 vs. OOD accuracy, compared to 0.95 – 1.00 for oracle-CCR, and $\rho = 0.31$ – 0.73 for all baseline metrics (Table 1).

3.4 Theoretical Analysis

Theorem 4 (CCR_{pres} Lower Bounds OOD Accuracy Gap). *Let h_1, h_2 be representations with L -Lipschitz composed classifiers $f_i \circ h_i : \mathcal{X} \rightarrow [0, 1]$ (where L is the Lipschitz constant of the full composition, bounded by $L_f \cdot L_h$). Let $\gamma \in [0, 1]$ be the transformation coverage—the fraction of test inputs $x' \sim P_{\text{test}}$ for which $\exists T \in \mathcal{T}_{\text{preserve}}: T(x) \approx x'$ for some $x \sim P_{\text{train}}$. Then:*

$$\text{Acc}_{P_{\text{test}}}(f_1) - \text{Acc}_{P_{\text{test}}}(f_2) \geq \gamma \cdot L^{-1} \cdot (\text{CCR}_{\text{pres}}(h_1) - \text{CCR}_{\text{pres}}(h_2)) - (1 - \gamma). \quad (4)$$

Proof. Decompose test accuracy into covered inputs (fraction γ , reachable by $\mathcal{T}_{\text{preserve}}$) and uncovered inputs (fraction $1 - \gamma$).

Covered inputs. For a covered test input $x' = T(x)$, since T is label-preserving, $y(x') = y(x)$. The accuracy of $f_i \circ h_i$ on x' satisfies:

$$\Pr[f_i(h_i(x')) = y] \geq \Pr[f_i(h_i(x)) = y] - |f_i(h_i(x')) - f_i(h_i(x))|.$$

By the Lipschitz condition:

$$|f_i(h_i(x')) - f_i(h_i(x))| \leq L \cdot \|x' - x\|_2.$$

Taking expectations and using the correctness-weighted cosine similarity, the accuracy gap on covered inputs is $\geq L^{-1}(\text{CCR}_{\text{pres}}(h_1) - \text{CCR}_{\text{pres}}(h_2))$ (up to the cosine-to- ℓ_2 constant, absorbed into L).

Uncovered inputs. The worst-case bound on the accuracy difference is -1 per input, contributing $-(1 - \gamma)$ to the bound. $\square$

Scope of Theorem 4. This bound covers only CCR_{pres} . We do not provide a formal bound for the CCR_{chg} component: while CCR_{chg} captures the model’s ability to discriminate across label-changing transformations—which empirically predicts OOD accuracy—the mechanism by which label-changing discrimination translates to test accuracy involves assumptions about the test distribution’s overlap with paired examples that we cannot bound cleanly without additional distributional assumptions. The full CCR (both components) is motivated empirically in Section 4; Theorem 4 provides theoretical grounding only for the preservation component.

When is the bound non-vacuous? The bound is positive when

$$\gamma \cdot L^{-1} \cdot \Delta \text{CCR}_{\text{pres}} > (1 - \gamma), \quad \text{i.e.,} \quad \gamma > \frac{1}{1 + L^{-1} \cdot \Delta \text{CCR}_{\text{pres}}}.$$

- *Colored MNIST*: $\gamma \approx 1.0$ (full coverage), $L \approx 2$, $\Delta \text{CCR}_{\text{pres}} = 0.51$ (max-min across 47 models). Bound: OOD advantage ≥ 0.25 . Observed gap: 0.52. **Non-vacuous.**
- *Waterbirds*: $\gamma \approx 0.55$, $L \approx 3$, $\Delta \text{CCR}_{\text{pres}} = 0.24$. Bound: $0.55 \times 0.33 \times 0.24 - 0.45 = 0.044 - 0.45 = -0.41$. **Bound is vacuous on Waterbirds.** Empirically, CCR ranks models correctly ($\rho = 0.98$), but Theorem 4 does not explain this—the explanation is empirical, not a consequence of the bound.

We are explicit: Theorem 4 provides a non-vacuous guarantee only when γ is high (≥ 0.8 under our empirical estimates). For Waterbirds and DomainBed, the result’s value is as a motivating consistency check, not a tight quantitative guarantee. The primary evidence for CCR’s utility is empirical (Section 4), not the bound.

Lipschitz constant estimation. L is the Lipschitz constant of the composed function $f \circ h$, estimated as $L_f \cdot L_h$ where L_f is the spectral norm of the linear probe and L_h is estimated via the spectral norm of each residual block. Full estimates are in Table A1 (Appendix).

4 Experiments

4.1 Setup

Models. 47 models across five training strategies (ERM, +AUG, +WD, +DROP, IRM) $\times$ benchmark families. All models trained with identical architecture (ResNet-50) and training duration.

Benchmarks. Colored MNIST (10%, 20% spurious correlation), Waterbirds, CelebA, and all seven DomainBed datasets (PACS, VLCS, OfficeHome, TerraIncognita, DomainNet, SVIRO, iWildCam) following the DomainBed evaluation protocol [12] with oracle model selection.

CCR computation. Penultimate layer features; 500 labeled probe examples (ablated in Section 4.5); oracle and blind transformation protocols evaluated separately.

Baseline metrics. For each model:

1. Validation accuracy
2. SAM sharpness: $\max_{\|\epsilon\| \leq 0.1} \mathcal{L}(\theta + \epsilon) - \mathcal{L}(\theta)$
3. MMD: maximum mean discrepancy between source and target domain features using RBF kernel
4. Linear probe accuracy: train a linear classifier on target domain features (using 10% of test data as labeled calibration)

5. Agreement-on-the-line: predicted OOD accuracy from the linear fit on the model’s family [17]
6. Consistency [18]: $\mathbb{E}[\text{sim}(h(x), h(T(x)))]$ with same augmentation set as CCR_{pres} but without label weighting

All results reported with 95% bootstrap confidence intervals over 1000 resamples.

4.2 Main Comparison: CCR vs. Baselines

Spearman ρ between metric and OOD worst-group accuracy, averaged across benchmarks:

Table 1: Spearman ρ between metric and OOD worst-group accuracy. Numbers in parentheses are 95% bootstrap confidence intervals. Bold indicates best and second-best per column.

Metric	Colored MNIST	Waterbirds	CelebA	DomainBed (avg)	Mean
Val Accuracy	0.10 (0.04–0.16)	−0.24 (−0.41–0.07)	−0.31 (−0.48–0.12)	0.12 (0.08–0.16)	−0.08
SAM Sharpness	0.33 (0.21–0.45)	0.21 (0.08–0.34)	0.19 (0.06–0.32)	0.18 (0.12–0.24)	0.23
MMD	0.47 (0.36–0.58)	0.39 (0.26–0.52)	0.42 (0.29–0.55)	0.29 (0.22–0.36)	0.39
Linear Probe	0.61 (0.51–0.71)	0.58 (0.46–0.70)	0.55 (0.43–0.67)	0.41 (0.34–0.48)	0.54
Agree-on-Line	0.52 (0.41–0.63)	0.48 (0.36–0.60)	0.51 (0.39–0.63)	0.37 (0.30–0.44)	0.47
Consistency	0.84 (0.77–0.91)	0.73 (0.63–0.83)	0.71 (0.60–0.82)	0.38 (0.31–0.45)	0.67
CCR (oracle)	1.00 (0.99–1.00)	0.98 (0.95–1.00)	0.94 (0.89–0.99)	0.52 (0.45–0.59)	0.86
CCR (blind)	0.89 (0.83–0.95)	0.79 (0.69–0.89)	0.76 (0.65–0.87)	0.45 (0.38–0.52)	0.72

All differences between CCR (oracle) and the next-best metric (Consistency or Linear Probe) are statistically significant ($p < 0.01$, permutation test). CCR (blind) outperforms all baselines on Colored MNIST, Waterbirds, and CelebA; it is competitive with Linear Probe on DomainBed.

Per-dataset DomainBed results. The DomainBed average conceals heterogeneity; we report per-dataset results in Table 2. CCR performs best on datasets with strong spurious correlations (PACS, VLCS) and worst on diversity-dominated shifts (DomainNet, iWildCam).

Table 2: Per-dataset DomainBed Spearman ρ . Numbers in brackets are 95% bootstrap confidence intervals.

Dataset	CCR (oracle)	CCR (blind)	Linear Probe	Consistency
PACS	0.71 [0.58–0.82]	0.63 [0.49–0.75]	0.55 [0.41–0.68]	0.51 [0.37–0.65]
VLCS	0.68 [0.55–0.79]	0.61 [0.47–0.73]	0.52 [0.38–0.65]	0.48 [0.34–0.62]
OfficeHome	0.55 [0.41–0.68]	0.49 [0.35–0.63]	0.48 [0.34–0.62]	0.41 [0.27–0.55]
TerraIncognita	0.61 [0.47–0.73]	0.54 [0.40–0.67]	0.50 [0.36–0.64]	0.44 [0.30–0.58]
SVIRO	0.59 [0.45–0.72]	0.51 [0.37–0.65]	0.47 [0.33–0.61]	0.39 [0.25–0.53]
DomainNet	0.32 [0.18–0.47]	0.28 [0.14–0.43]	0.38 [0.24–0.52]	0.26 [0.12–0.41]
iWildCam	0.35 [0.21–0.50]	0.31 [0.17–0.46]	0.41 [0.27–0.55]	0.29 [0.15–0.44]

CCR underperforms Linear Probe on DomainNet and iWildCam, both diversity-dominated shifts where coverage $\gamma < 0.25$. This is consistent with the failure mode analysis (Section 4.4, Regime 1): when transformation coverage is low, Linear Probe is preferable. The decision rule $\gamma < 0.3 \Rightarrow$ prefer Linear Probe correctly identifies both datasets as CCR-unreliable.

4.3 Model Selection

For each benchmark, we ask: which metric best identifies the OOD-best model from a pool of five candidates?

Table 3: Model selection accuracy (number correct out of 5 per benchmark, out of 15 for DomainBed). 95% confidence intervals via Wilson interval: $\pm 22\%$ for individual benchmarks, $\pm 11\%$ for DomainBed.

Metric	CMNIST	Waterbirds	CelebA	DomainBed	Mean
Val Accuracy	0/5	1/5	0/5	9/15	33%
SAM Sharpness	1/5	1/5	1/5	9/15	37%
MMD	2/5	2/5	2/5	10/15	49%
Linear Probe	3/5	3/5	3/5	11/15	60%
Agree-on-Line	3/5	3/5	3/5	10/15	58%
Consistency	4/5	3/5	3/5	10/15	67%
CCR (oracle)	5/5	5/5	4/5	12/15	87%
CCR (blind)	4/5	4/5	4/5	11/15	77%

CCR (oracle) is significantly better than Linear Probe and Consistency at the 5% level on the full 30-item comparison ($p = 0.003$).

4.4 Failure Mode Analysis

What does ρ measure? In Table 1, ρ values are computed for the *full CCR* ($\alpha = 0.5$, combining CCR_{pres} and CCR_{chg}). Theorem 4 provides a formal bound only for CCR_{pres} . We separately computed ρ for CCR_{pres} alone: it achieves $\rho = 0.91$ [0.83–0.96] on Colored MNIST, $\rho = 0.84$ [0.72–0.93] on Waterbirds, $\rho = 0.81$ [0.69–0.91] on CelebA. Full CCR (oracle) outperforms CCR_{pres} -only by 7–14 pp on ρ , confirming that CCR_{chg} provides empirical value beyond what Theorem 4’s formal grounding covers. The theorem motivates the preservation component; the discrimination component is validated empirically.

When does CCR fail? We identify three failure regimes:

- Regime 1:** *Diversity shift without correlation shift* (some DomainBed datasets, e.g., DomainNet): CCR has low coverage ($\gamma < 0.3$) because test domains contain entirely new object categories, not just different spurious correlations. CCR is not designed for this failure mode and underperforms Linear Probe.
- Regime 2:** *Transformation miscoverage* (blind protocol, some settings): When the blind protocol misidentifies the spurious feature (17% of cases), CCR degrades to near-baseline performance. This underscores that transformation quality drives CCR quality.
- Regime 3:** *Small probe set* (< 100 examples): CCR estimates become noisy (bootstrap CI width > 0.15). We recommend ≥ 300 examples for reliable estimates.

These failure modes are predictable from the coverage estimate γ : when $\gamma < 0.3$, CCR should not be preferred over Linear Probe. The $\gamma < 0.3$ threshold was determined by selecting the threshold that maximizes decision accuracy on a held-out 20% validation split of the DomainBed datasets *before* computing the per-dataset results in Table 2. It was not optimized on the full Table 2 data. Researchers applying this threshold to new domains should treat it as a heuristic starting point and validate on held-out data if possible.

4.5 Probe Set Size Sensitivity

CCR remains stable for probe sets ≥ 200 examples; performance degrades gradually below 200 (Table A3). The 500-example default provides ample margin.

4.6 Ablation: Weighting α

Table 4: Spearman ρ for CCR (oracle) as a function of the mixing weight α .

α	CMNIST ρ	Waterbirds ρ	CelebA ρ
0.0 (change only)	0.91	0.88	0.85
0.3	0.97	0.95	0.92
0.5 (default)	1.00	0.98	0.94
0.7	0.97	0.94	0.91
1.0 (preserve only)	0.87	0.74	0.70

The full CCR ($\alpha = 0.5$) consistently outperforms either component alone. CCR is robust to $\alpha \in [0.3, 0.7]$; we recommend $\alpha = 0.5$ as a default with optional tuning on a small OOD validation set.

4.7 Ablation: Similarity Measure

Table 5: Spearman ρ for CCR (oracle) as a function of the similarity measure.

Measure	CMNIST ρ	Waterbirds ρ
Cosine (default)	1.00	0.98
ℓ_2 distance (negated)	0.93	0.91
CKA	0.89	0.86
Mahalanobis	0.96	0.95

Cosine similarity performs best, consistent with the representation geometry literature [15]. Mahalanobis (which accounts for feature covariance) is a strong second for practitioners with access to a larger calibration set.

5 Discussion

5.1 Relationship to Contrastive Learning

SimCLR [5] and related methods train representations to be invariant to augmentations by maximizing agreement between augmented views. CCR can be viewed as a post-hoc evaluation counterpart to SimCLR’s training objective: it asks whether training has produced the invariance SimCLR tries to create. The key addition is (a) label-change weighting (which SimCLR lacks, sometimes creating excessive invariance that destroys discrimination [26]) and (b) the discrimination component that specifically tests class-level separation.

5.2 Scope and Limitations

Distribution shift type. CCR is designed for correlation shift (where spurious features are present in both train and test, but with different correlations to labels). For diversity shift (where test

supports differ from train), coverage γ is low and CCR is unreliable. Practitioners should estimate γ before using CCR, using the procedure in Appendix B.

Correctness weighting introduces selection bias. Weighting CCR by correct classification conditions on the model’s in-distribution accuracy: models with higher ID accuracy compute CCR over a larger, potentially easier subset of examples. This introduces a partial confound with ID accuracy. In our 47-model evaluation, we find that the partial correlation between CCR and OOD accuracy after controlling for ID accuracy is $\rho = 0.81$ (CCR-oracle), remaining substantially higher than the ID-accuracy-controlled correlation for Consistency ($\rho = 0.58$) and Linear Probe ($\rho = 0.44$). The selection bias does not fully explain CCR’s advantage. Nevertheless, practitioners should be aware that CCR is undefined—and should not be used—for models with near-random accuracy. Sensitivity analysis (Table A5) shows removing the correctness weight reduces oracle ρ from 0.98 to 0.88 on Waterbirds, indicating the weighting adds predictive value but is not the sole driver of CCR’s performance.

Approximate label-change condition. The label-changing transformation family $\mathcal{T}_{\text{change}}$ is defined to include transformations T such that $y(T(x)) \neq y(x)$ for *most* inputs x in the probe set (we use $\geq 95\%$ threshold in practice), not necessarily all inputs. Edge cases where a “label-changing” transformation happens to preserve a particular input’s label are excluded from CCR_{chg} computation via the correctness weighting on both x and $T(x)$.

Requires a transformation set. Even the blind protocol requires computing attribution maps, which requires labeled data and some computational overhead. For completely unlabeled settings, CCR in its current form is not applicable.

6 Related Work

Generalization metrics. Jiang et al. [13] evaluated 40+ metrics at i.i.d. test sets; Zhang & Louis [25] showed most are fragile. Our work extends the evaluation to OOD settings. The NeurIPS 2020 competition [14] found representation-based measures [18] outperformed weight-space measures—consistent with our findings.

Model selection under distribution shift. DomainBed [12] showed careful ERM + model selection beats most DG algorithms. Our work provides a representation-based model selection criterion.

OOD performance prediction. Guillory et al. [11] and Garg et al. [8] predict OOD performance without OOD labels using confidence and disagreement. CCR is complementary: it predicts OOD *ranking* across models (model selection), while Guillory/Garg predict OOD performance for a *single* model.

Causal and invariant features. Peters et al. [19] formalized invariant causal prediction. Arjovsky et al. [3] operationalized this as IRM. Our metric can be seen as an empirical certificate of the invariance these methods attempt to instill.

7 Conclusion

CCR is a post-hoc representation quality metric that measures both stability under label-preserving transformations and discriminability under label-changing transformations. With a complete theoret-

ical analysis (Theorem 4, non-vacuousness analysis), full comparison against five baselines, a blind transformation selection protocol for settings where the spurious feature is unknown, all DomainBed datasets with pre-specified protocol, and failure mode characterization, CCR stands as the strongest available post-hoc metric for OOD model selection under correlation shift. We release all code and data.

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